



INDIA



UAE



MALAYSIA



AUSTRALIA

PYTHON FULL STACK WITH AI



OUR BRANCHES: CHENNAI|COIMBATORE|BANGALORE|KANCHIPURAM|TRICHY|HOSUR|TIRUPPUR|PUNE|DUBAI|MALAYSIA

www.pumotechnovation.com

TEAM : IT
BRANCH : MARATHAHALLI - BANGALORE

Technologies	Sessions	Topics	• Subtopics	Student Sign	Staff Sign
HTML5	Session 1	HTML Basics	• What is HTML? • Why do we use it? • Structure of HTML documents • Tags, Element, Attributes, • <html>, <head>, • <body>		
	Session 2	Formatting	• Text formatting tags: • <i> <u> • • Anchor tag <a> • Internal & external links • target="_blank" • Email & phone link (mailto:, tel:)		
	Session 3	Images & Multimedia	• img tag, src, alt, height, width • Image hosting basics • Audio tag <audio> • Video tag <video> • Controls, autoplay, • loop		
	Session 4	List & Tables	• Ordered list • Unordered list • Description list <dl> • Table tags: • <table> <tr> <td> <th> • <thead> <tbody> • <tfoot> • Rowspan, Colspan		

	Session 5	Form (part 1)	•<form> basics •Input types: text, email, password, number, date, color, file •Labels, Placeholder •Buttons		
	Session 6	Forms (Part2)	•<select> <option> •<textarea> •Radio buttons & Checkboxes •Form validation attributes: required, minlength, maxlength, pattern •method="GET" vs POST" •action attribute		
	Session 7	Semantic Tags	•Why semantic tags? •<header> <nav> <section> •<article> <aside> <footer> •<figure> <figcaption> •SEO benefits		
CSS	Session 8	CSS Styles	•CSS Selectors & Properties •Element, class, id, universal selectors •Grouping, descendant, child, adjacent selectors •color, background-color, font-size, font-family, text-align, text-decoration, etc.		
	Session 9	Box Model	•Box Model •Content, padding,		

			border, margin • box-sizing: content-box vs border-box • overflow, outline, visibility, display basics		
	Session 10	Units	• CSS Units & Sizing • Absolute vs relative units (px, em, rem, %, vw, vh) • Width, height, min/max • Centering elements (text, block)		
	Session 11	Position	• CSS Positioning & Z-Index • static, relative, absolute, fixed, sticky • top, left, right, bottom • z-index and stacking context		
	Session 12	Control Layout	• Display, Visibility & float • display: block, inline, inline-block, none • visibility, float, and clear • overflow, clearfix method		
	Session 13	Flex	• Flexbox Layout • display: flex • Direction, justify-content, align-items • Flex-grow, shrink, basis • Flexbox responsive design examples		
	Session 14	Grid	• CSS Grid Layout • display: grid • Columns, rows, gaps, fr unit		

			• Grid-template, area, placement • Grid vs Flexbox comparison		
	Session 15	Advanced CSS	• Transitions, Transforms & Animations • CSS transition, transform (scale, rotate, translate) • Keyframe animations (@keyframes, animation-name, duration, iteration, delay) • Simple hover animations		
	Session 16	Responsive	• Media Queries • Responsive design intro • @media rules: breakpoints for mobile, tablet, desktop • Practice Project: Responsive Landing Page (with flex/grid, form, animation)		
Bootstrap	Session 17	Bootstrap Fundamentals	• Intro Bootstrap • Bootstrap CDN • File Structure		
	Session 18	Grid System	• (Container, Row, Column) • Breakpoints & Responsive Basics		
	Session 19	Layout & Utilities	• Typography & Text Utilities • Colors & Background Utilities • Spacing (Margin &		

			Padding) • Display, Flex & Position Utilities		
	Session 20	Components – Part 1	• Buttons & Button Groups • Cards & Card Layouts • Images & Image Utilities • Tables & Responsive Tables		
	Session 21	Components – Part 2	• Navbar & Navigation • Forms & Form Controls • Input Groups & Validation • List Groups & Badges		
	Session 22	Responsive Design & Mini Projects	• Responsive Design • Grid-Based Page Layouts • Bootstrap Icons • Theme Customization Basics • Mini Project (Landing Page / Admin Dashboard / Portfolio)		
	Session 23	Introduction to JavaScript	• What is JavaScript? • JavaScript vs HTML vs CSS • Where to write JS (inline, internal, external) • Console & developer tools • First JavaScript program		
	Session 24	Variables & Data Types	• var, let, const • Data types (number, string, boolean, null, undefined)		

JavaScript			•Type conversion •typeof operator		
	Session 25	Operators & Expressions	•Arithmetic operators •Assignment operators •Comparison operators •Logical operators •Ternary operator		
	Session 26	Control Statements	•if, else, else if •switch •Conditional logic practice		
	Session 27	Loops	•for, while, do...while •break, continue •Loop patterns & exercises		
Core JavaScript Concepts	Session 28	Functions	•Function declaration & expression •Parameters & arguments •Return statement •Arrow functions		
	Session 29	Arrays	•Creating arrays •Array methods (push, pop, shift, unshift) •Looping arrays •Array basics exercises		
	Session 30	Strings & String Methods	•String properties •Methods (length, slice, replace, toUpperCase) •Template literals		
	Session 31	Objects	•Object creation •Properties & methods •this keyword •Object iteration		

	Session 32	Math & Date Objects	• Math methods • Random number generation • Date & time handling		
Advanced JavaScript Basics	Session 33	Scope & Hoisting	• Global & local scope • Block scope • Hoisting concept		
	Session 34	Callbacks & Higher Order Functions	• Callback functions • map, filter, reduce • Practical examples		
	Session 35	Error Handling	• try, catch, finally • throw • Common JS errors		
	Session 36	DOM Introduction	• What is DOM? • Selecting elements • Changing content & styles		
	Session 37	DOM Events	• Event types • addEventListener • Event object • Form & mouse events		
DOM & Browser APIs	Session 38	DOM Manipulation	• Create & remove elements • Class & attribute manipulation • Traversing DOM		
	Session 39	Forms & Validation	• Form handling • Input validation • Prevent default behavior		
	Session 40	Browser Storage	• localStorage • sessionStorage • Cookies basics		

	Session 41	Timing Functions	• setTimeout • setInterval • Clear timers		
	Session 42	BOM & Window Object	• Window methods • Screen, location, history • Popup handling		
Modern JavaScript (ES6+)	Session 43	ES6 Features	• let vs const • Arrow functions • Template strings • Destructuring		
	Session 44	Spread & Rest Operators	• Spread with arrays & objects • Rest parameters • Practical examples		
	Session 45	Modules	• import & export • Module structure • Use cases		
	Session 46	Asynchronous JavaScript	• Synchronous vs asynchronous • Callbacks vs Promises • async & await		
	Session 47	Fetch API	• Fetching data from APIs • JSON handling • Error handling in fetch		
Develop	Session 48	OOP in JavaScript	• Classes & objects • Constructor • Inheritance		
	Session 49	Event Loop & JS Internals	• Call stack • Event loop • Microtasks vs macrotasks		
	Session 50	Performance & Best Practices	• Clean code practices • Optimization basics		

Projects			• Debugging techniques		
	Session 51	JavaScript Interview Preparation	• Common interview questions • Coding challenges • Output-based questions		
	Session 52	Mini Projects	• To-Do App • Calculator • Form Validation App • API Data Fetch App		
React.js	Session 53	React Fundamentals	• What is React? • SPA vs MPA • React features & advantages • React vs Angular vs Vue • Node.js & npm overview		
	Session 54	Environment Setup	• Node.js installation • create-react-app • Vite introduction • Project folder structure • Running first React app		
	Session 55	JSX & Rendering	• JSX syntax rules • Expressions in JSX • ReactDOM rendering • Fragment usage • JSX best practices		
	Session 56	Components Basics	• Function components • Component naming rules • Reusability • Component folder structure		

	Session 57	Props (Component Communication)	• What are props? • Passing data to components • Props validation (PropTypes) • Default props		
React Hooks	Session 58	State & useState Hook	• What is the state? • useState() hook • Updating state • Multiple states • Controlled components		
Rendering	Session 59	Events & Conditional Rendering	• Handling events • Passing arguments to events • Conditional rendering techniques • Logical AND & ternary operator		
	Session 60	Lists & Keys	• Rendering lists • map() in JSX • Keys & their importance • Dynamic UI rendering		
	Session 61	Styling in React	• Inline styling • CSS files • CSS Modules • Styled-components (overview)		
	Session 62	Forms & Validation	• Controlled forms • Form submission • Input handling • Basic form validation		

Validation	Session 63	useEffect Hook	• Side effects in React • useEffect() lifecycle • Dependency array • Cleanup function		
API	Session 64	API Integration	• Fetch API in React • Axios introduction • Loading & error states • Displaying API data		
Router	Session 65	React Router	• SPA routing concept • React Router setup • Routes, Route, Link • useParams, useNavigate		
	Session 66	Lifting State & Context API	• Lifting state up • Prop drilling problem • Context API • useContext() hook		
	Session 67	Advanced Hooks	• useRef • useMemo • useCallback • Performance optimization basics		
	Session 68	Component Lifecycle & Optimization	• React component lifecycle • Re-rendering concepts • Memoization (React.memo) • Best practices		

React Component Cycle	Session 69	Error Handling & Debugging	• Error boundaries • Common React errors • Debugging with DevTools • Console & breakpoints		
	Session 70	State Management Basics	• When state management is needed • Redux overview • Redux Toolkit basics • Store, reducer, action concepts		
Projects & Interview Prep	Session 71	Projects	• To-Do App • API-based Product List • Login & Register UI • Dashboard layout		
	Session 72	Introduction to Databases	• What is a database? • DBMS vs RDBMS • File system vs database • Types of databases (Relational, NoSQL, Cloud) • Real-world use cases		
	Session 73	Database Architecture	• Client-Server architecture • Tables, rows, columns • Primary key concept • Database schemas • Popular databases overview		

Database	Session 74	SQL Basics	• What is SQL? • SQL commands categories (DDL, DML, DQL, DCL, TCL) • CREATE DATABASE • CREATE TABLE • DROP, TRUNCATE		
	Session 75	Insert & Select	• INSERT INTO • SELECT * • Selecting specific columns • DISTINCT • LIMIT / TOP		
	Session 76	WHERE & Filtering	• WHERE clause • Comparison operators • BETWEEN • IN • LIKE • IS NULL		
	Session 77	Sorting & Aggregation	• ORDER BY • Aggregate functions • COUNT, SUM, AVG, MIN, MAX • GROUP BY • HAVING		
	Session 78	Joins	• INNER JOIN • LEFT JOIN • RIGHT JOIN • FULL JOIN • Join with multiple tables		

	Session 89	Subqueries	• Subquery basics • Nested subqueries • Subqueries in SELECT, WHERE, FROM • Correlated subqueries		
	Session 80	Constraints	• PRIMARY KEY • FOREIGN KEY • UNIQUE • NOT NULL • CHECK • DEFAULT		
	Session 81	Update & Delete	• UPDATE • DELETE • Safe update mode • Rollback concept		
Advanced SQL Concepts	Session 82	Functions	• String functions • Numeric functions • Date & time functions • User-defined functions (intro)		
	Session 83	Indexes	• What is indexing? • Types of indexes • Create & drop index • Index performance impact		
	Session 84	Views	• What is a view? • Create & drop view • Updatable views • Real-world use cases		
	Session 85	Stored Procedures	• Stored procedure basics • Parameters (IN, OUT) • Calling procedures • Advantages		

	Session 86	Triggers	•What is trigger? •BEFORE & AFTER triggers •INSERT, UPDATE, DELETE triggers •Use cases		
Transaction & Optimization	Session 87	Transactions	•ACID properties •START TRANSACTION •COMMIT •ROLLBACK •Savepoints		
	Session 88	Normalization	•Data redundancy •1NF, 2NF, 3NF •Boyce–Codd Normal Form •Denormalization		
	Session 89	Performance Tuning	•Query optimization •Explain plan •Index tuning •Common slow query issues		
	Session 90	Security & Access Control	•Users & roles •GRANT & REVOKE •SQL injection basics •Data protection		
	Session 91	Backup & Recovery	•Database backup types •Restore process •Import & export •Disaster recovery basics		
Developer	Session 92	Database Design	•ER diagrams •Relationships •Cardinality •Schema design		

Integration & Advanced Topics	Session 93	SQL with Applications	• Connecting DB with Python • CRUD operations • Prepared statements		
	Session 94	Advanced Joins & Window Functions	• Self join • Cross join • Window functions (ROW_NUMBER, RANK) • Analytical queries		
	Session 95	NoSQL Introduction	• What is NoSQL? • MongoDB basics • SQL vs NoSQL • Use cases		
	Session 96	Cloud Databases	• AWS RDS overview • Scaling concepts		
	Session 97	Case Studies	• E-commerce database design • Banking system schema • Student management system		
	Session 98	Complex SQL Queries	• Nested joins • Performance-based queries • Real-time query challenges		
	Session 99	Database Testing	• Data validation • Load testing basics • Data integrity checks		
	Session 100	Final Projects	• Complete database project • Schema + data + queries • Application integration demo		

Core Python With AI	Session 101	Python Introduction & Setup	• What is Python? • Python features & applications • Python vs other languages • Installing Python & IDE (VS Code / PyCharm) • Python execution flow		
	Session 102		• Interpreter vs compiler • REPL • Python file structure (.py) • First Python program		
	Session 103	Variables & Keywords	• Variables & naming rules • Python keywords • Data types overview		
	Session 104	Input, Output & Type Casting	• input() function • print() formatting • f-strings • Type casting (int(), float(), str()) • Multiple inputs • AI relevance: data preprocessing basics		
	Session 105	Operators	• Arithmetic operators(+, -, *, /) • Assignment operators(=) • Comparison operators(==, <, >) • Logical operators(and, or) • Bitwise operators • Membership & identity operators		

	Session 106	Control Flow (Decision Making)	• if, else, elif • Nested conditions • Short-hand if • Real-world condition logic (AI rules)		
	Session 107	Loops	• for loop • while loop • Sub Topics • break, continue, pass • Loop patterns • AI logic iteration examples		
	Session 108	Strings	• String creation & indexing • Slicing • String methods • String immutability • NLP string preprocessing intro		
	Session 109	Lists	• List creation • Access & modification • List methods • Nested lists • List comprehension • AI dataset list handling		
	Session 110	Tuples & Sets	• Tuple basics • Set basics • Tuple immutability • Set operations • Removing duplicates (AI data cleaning)		
	Session 111	Dictionary	• Dictionary creation • Key-value access • Dictionary methods • Nested dictionaries • JSON-like data handling		

			(AI APIs)		
	Session 112	Functions	• Function definition • Function call • Parameters & arguments • Return statement • Function reuse in AI logic		
	Session 113	Advanced Functions	• Default arguments • Keyword arguments • *args, **kwargs • Lambda functions • Functional programming basics		
	Session 114	Modules & Packages	• Importing modules • Built-in modules • math, random, datetime • Creating custom modules • AI library structure intro		
	Session 115	File Handling	• File modes • Reading & writing files • Text vs binary files • CSV handling • AI dataset file usage		
	Session 116	Exception Handling	• Errors vs exceptions • try, except, else, finally • Custom exceptions • AI pipeline error handling		
	Session 117	Object Oriented Programming	• Class & object • Constructor • Instance variables • Methods		

			• AI model class structure intro		
	Session 118	OOP Advanced	• Inheritance • Polymorphism • Encapsulation • Abstraction • Real-world AI class design		
	Session 119	Data Structures & Algorithms Basics	• Stack • Queue • Searching algorithms • Sorting algorithms • AI efficiency concept		
	Session 120	Python Standard Libraries	• os • sys • re (regex) • json • AI data parsing		
	Session 121	Database Connectivity	• Python with MySQL / SQLite • CRUD operations • SQL execution from Python • AI data storage basics		
	Session 122	AI Basics with Python	• What is AI? • AI vs ML vs DL • AI workflow • Python role in AI • Simple AI examples		
	Session 123	Machine Learning Basics	• Supervised learning • Unsupervised learning • ML algorithms overview • Scikit-learn intro		
	Session 124	Simple ML Model Implementation	• Dataset loading • Train-test split		

			• Model training • Prediction basics		
Django Framework	Session 125	Django Introduction	• What is Django? • MVT architecture • Django vs Flask vs FastAPI • Web application flow • Frontend vs Backend • HTTP request & response • Django project structure overview		
	Session 126	Environment Setup & First Project	• Virtual environment • Django installation • Creating project & app • django-admin startproject • python manage.py startapp • settings.py overview • Running development server		
	Session 127	URL Routing & Views	• URL dispatcher • Views basics • Function-based views • URL patterns • HttpResponse • App-level vs project-level URLs		
	Session 128	Templates & Static Files	• Django template engine • Template syntax • Template inheritance • Static files (CSS, JS, images) • Base layout creation		

	Session 129	Models & Database Basics	• ORM introduction • Model creation • Field types • Migrations • SQLite / MySQL setup • Model relationships intro		
	Session 130	Django Admin Panel	• Admin configuration • Superuser creation • Registering models • Customizing admin interface • Admin filters & search		
	Session 131	CRUD Operations	• Create & Read data • ORM queries • Displaying data in templates • Context data • Dynamic pages		
	Session 132	Update & Delete	• Update operations • Delete operations • Form-based update • Delete confirmation • Redirects • Messages framework		
	Session 133	Forms & Validation	• Django forms • Form classes • ModelForms • Form validation • CSRF protection		
	Session 134	Authentication & Authorization	• User authentication • Login & logout • User registration • Permissions • Role-based access		

	Session 135	Class Based Views (CBV)	• CBV basics • ListView • DetailView • CreateView • UpdateView • DeleteView		
	Session 136	REST API with Django REST Framework	• DRF introduction • Serializers • API views • CRUD APIs • JSON response		
	Session 137	Advanced DRF	• API security • Token authentication • JWT overview • API permissions • Pagination & filtering		
	Session 138	AI Integration Basics	• AI in Django apps • AI/ML workflow • Loading ML models in Django • Predict endpoint design • Use cases (recommendation, prediction)		
	Session 139	AI Model Integration	• Integrating trained AI models • Pickle / joblib models • Prediction views • Handling input data • Displaying AI results		
	Session 140	AI APIs Integration	• External AI services • REST API consumption • OpenAI / HuggingFace (concept) • Chatbot integration • NLP / text processing		

			flow		
	Session 141	Background Tasks & Performance	• Asynchronous tasks • Celery overview • Background AI processing • Caching basics • Performance optimization		
	Session 142	Deployment & Security	• Production deployment • Environment variables • Static & media handling • Deployment (AWS / Render / Railway) • Security best practices		
	Session 143	Real-World Project	• AI-Powered Django Application • AI Resume Analyzer • AI Student Performance Predictor • AI Chatbot Web App • Recommendation System		
	Session 144	AWS & Cloud Fundamentals	• What is Cloud Computing? • Cloud service models (IaaS, PaaS, SaaS) • Why AWS? • AWS global infrastructure • Regions • Availability Zones • Edge locations • AWS use cases (hosting, backup, AI, DevOps)		

AWS Hosting	Session 145	AWS Account Setup & IAM	• Creating AWS Free Tier account • AWS Management Console overview • Root user vs IAM user • IAM users, roles & policies • Security best practices • Billing dashboard overview		
	Session 146	Choosing AWS Services for Hosting	• AWS compute services overview • AWS storage services overview • EC2 vs Lightsail • S3 vs EBS vs EFS • When to use which service • Architecture for simple web hosting		
	Session 147	EC2 (Launch & Configuration)	• What is EC2? • EC2 instance types • AMI selection • Key pair creation • Security groups • Launching EC2 (Linux) • Connecting via SSH / EC2 Connect		
	Session 148	Hosting Website on EC2	• Installing web server • Apache / Nginx installation • Hosting HTML / Django app • Opening ports (HTTP/HTTPS) • Public IP & Elastic IP • Basic troubleshooting		

	Session 149	S3 Bucket	• What is Amazon S3? • Bucket creation • Uploading files • Bucket permissions • Public access settings • Static website hosting • S3 vs EC2 hosting comparison		
	Session 150	Real-Time Hosting Flow	• Complete hosting architecture • EC2 + S3 architecture • Domain & Route 53 (overview) • SSL & HTTPS (concept) • Cost optimization tips • AWS Free Tier limits • Hosting checklist		
Version	Session 151	Version Control & Git Basics	• What is a Version Control System? • Centralized vs Distributed VCS • Why Git? • Git architecture • Working directory • Staging area • Repository • Git workflow • Git vs GitHub		
	Session 152	Git Installation & Configuration	• Installing Git (Windows / Linux / Mac) • Git Bash overview • git --version • git config --global user.name • git config --global user.email • Global vs local config • .gitignore file		

Control System	Session 153	GitHub Setup & Repositories	• Creating GitHub account • GitHub interface overview • Creating repositories • Local repo to GitHub • HTTPS vs SSH • Cloning repositories • README.md basics		
	Session 154	Git Basic Commands & Commits	• Git basic commands • git init • git status • git add • git commit • git log • git diff • Commit		
	Session 155	Branching, Merging & Collaboration	• Branching concept • Merging strategies • git branch • git checkout / git switch • git merge • Merge conflicts • Pull requests • Team collaboration workflow		
Capstone Project	Session 156	(Django + AI Project Final Implementation)	• Capstone project objective • Real-world problem statement • System architecture diagram • Data flow explanation		

Foundations of AI & Prompt Engineering	Session 157	Introduction to Prompt Engineering	• What is Generative AI How LLMs work • Prompt basics • Zero-shot prompting • Few-shot prompting • Role-based prompts • Output formatting		
	Session 158	Structured Prompt Writing	• Role definition • Context setting • Clear instructions • Constraints • Output structure • Error reduction • Prompt refinement		
	Session 159	Prompting for Python Programming	• Function generation • OOP code prompts • Exception handling • Code explanation • Code refactoring • Debug prompts • Test case generation		
Foundations of AI & Prompt Engineering	Session 160	Prompting for Backend	• API generation • CRUD operation • Model creation • Authentication logic • JWT prompts • Middleware generation • Error handling		
	Session 161	Prompting for Databases	• SQL query generation • Query optimization • Schema design • Migration scripts		

			• ORM usage • Index suggestion • Debugging SQL errors		
Foundations of AI & Prompt Engineering	Session 162	Prompting for Frontend	• UI component generation • Form creation • Validation logic • State management • API integration • Responsive design • Debug UI issues		
	Session 163	Debugging & Optimization	• Stack trace analysis • Performance tuning • Memory issues • Code review prompts • Security checks • Vulnerability detection • Refactor suggestions		
	Session 164	Testing & Automation	• Unit test generation • Pytest prompts • Mocking APIs • Integration testing • Test coverage • CI script prompts • Bug reproduction		